

British Math Olympiad Round 1 1979 — P3/6

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June 12, 2026

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S is a set of distinct positive odd integers $(a_i)_{i=1}^n$. No two differences $|a_i - a_j|$ are equal, $1 \leq i < j \leq n$.

Prove that $a_1 + a_2 + \cdots + a_n \geq \frac{1}{3}n(n^2 + 2)$.

Solution

Without loss of generality, order the numbers such that $a_1 < a_2 < \cdots < a_n$. Note that since S is comprised only of odd distinct integers, all the differences $|a_i - a_j|$ are even. So for some pairwise distinct positive even integers $d_2, d_3, \dots, d_n$ we know that

$$a_k = a_{k-1} + d_k, \quad (k = 2, 3, \dots, n).$$

Observe that for some integer $m \geq 2$.

$$\begin{aligned} \sum_{k=2}^m (a_k) &= \sum_{k=2}^m (a_{k-1} + d_k) \\ &= \sum_{k=1}^{m-1} (a_k) + \sum_{k=2}^m (d_k) \\ a_m - a_1 &= \sum_{k=2}^m (d_k). \end{aligned}$$

Since we want to minimise a_m we should let $d_2, d_3, \dots, d_m$ just be the first $(m-1)$ positive even integers. These sum to $2(1 + 2 + \cdots + (m-1)) = (m-1)m$. Thus

$$a_m \geq a_1 + m(m-1).$$

Now clearly $a_1 \geq 1$ so we deduce that $a_m \geq m^2 - m + 1$. Thus

$$\begin{aligned} a_1 + a_2 + \cdots + a_n &\geq 1 + \sum_{k=2}^n (k^2 - k + 1) \\ &= 1 + \left[\frac{n(n+1)(2n+1)}{6} - 1 \right] - \left[\frac{n(n+1)}{2} - 1 \right] + (n-1) \\ &= \frac{1}{3}n(n^2 + 2). \end{aligned}$$

which is the desired result. ■